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Datacommunicatie en netwerken

Bachelor in de elektromechanica – automatisering

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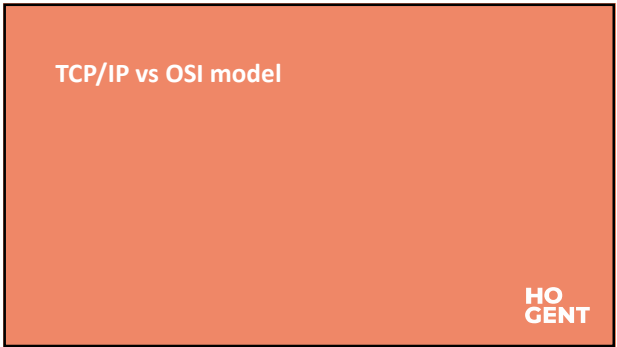


"I have not failed. I've just found 10,000 ways that won't work."

~Thomas A. Edison



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Inleiding – OSI model

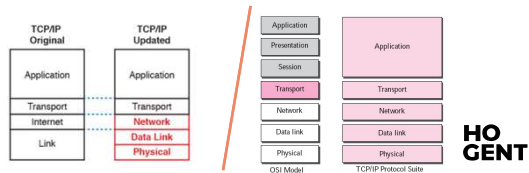
- OSI = Open Systems Interconnection (1979)
 - Set van afspraken ontwikkeld om communicatie tussen verschillende hardware mogelijk te maken => lagenmodel



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Inleiding – TCP/IP model

- Vereenvoudigd OSI model
 - Applicatie laag neemt Application, Presentation en Session over
- OSI in de praktijk, praktisch ieder netwerkkapparaat ondersteunt TCP/IP

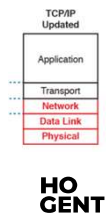


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Inleiding – TCP/IP model

De werking van een computernetwerk begrijpen
=
Het TCP/IP model begrijpen

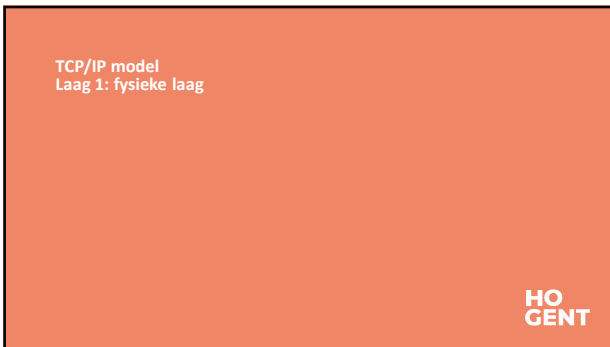
- We bespreken systematisch de verschillende lagen ☺
- Please Do Not Take Anything (TCP/IP)
- Please Do Not Take Something Precious Away (OSI)



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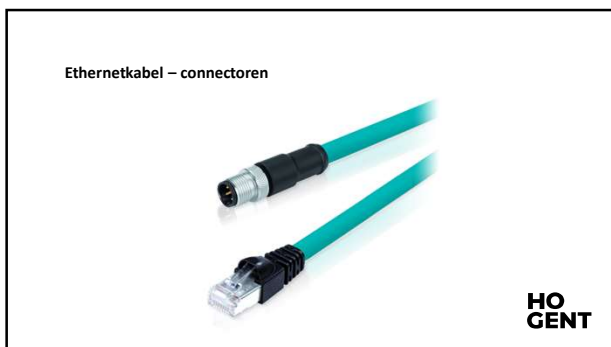
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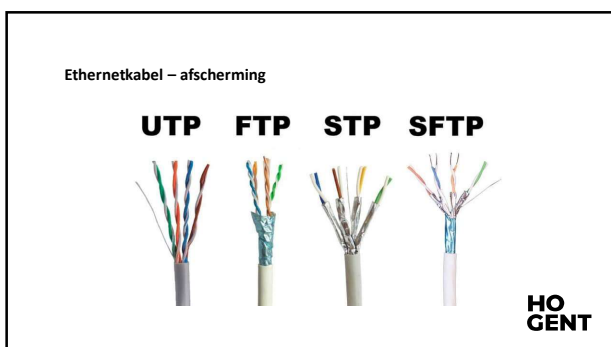
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Ethernetkabel – categorie

Oefening: ga op onderzoek en vul de tabel aan.

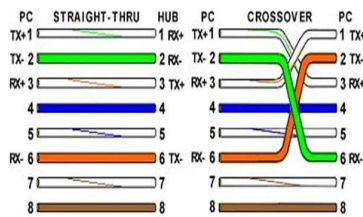
Categorie	+/- 30m	+/- 50m	100m	Connector
cat5e				
cat6				
cat6a				
cat7				
cat8				

Hoe kan je zien als een kabel cat5e, cat6, cat6a, cat7 of cat8 is?
Wat als je verder moet dan 100m?
Wat in de industrie?

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Ethernetkabel – straight-thru en crossover



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Oefening RJ-45 vs M12



M12 versus RJ45 Ethernet connection systems

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Abstract

Often it seems to be a matter of course, but people could argue against the superiority of the M12 connection system over the well-known RJ45 connection and wonder if a connection system designed to do no more than build up a redundant connection. Industrial Ethernet calling requires connection solutions that are simply not available.

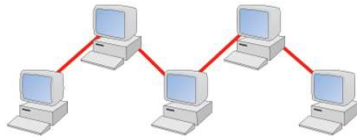
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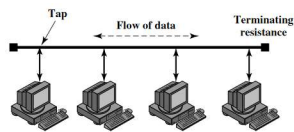
Line topologie



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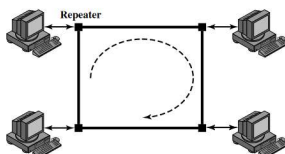
Bus topologie



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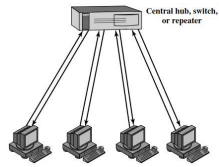
Ring topologie



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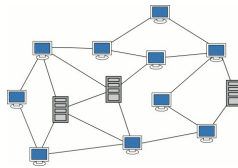
Star topologie



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Mesh topologie



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<https://www.youtube.com/watch?v=v1JEuzBkOD8>

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